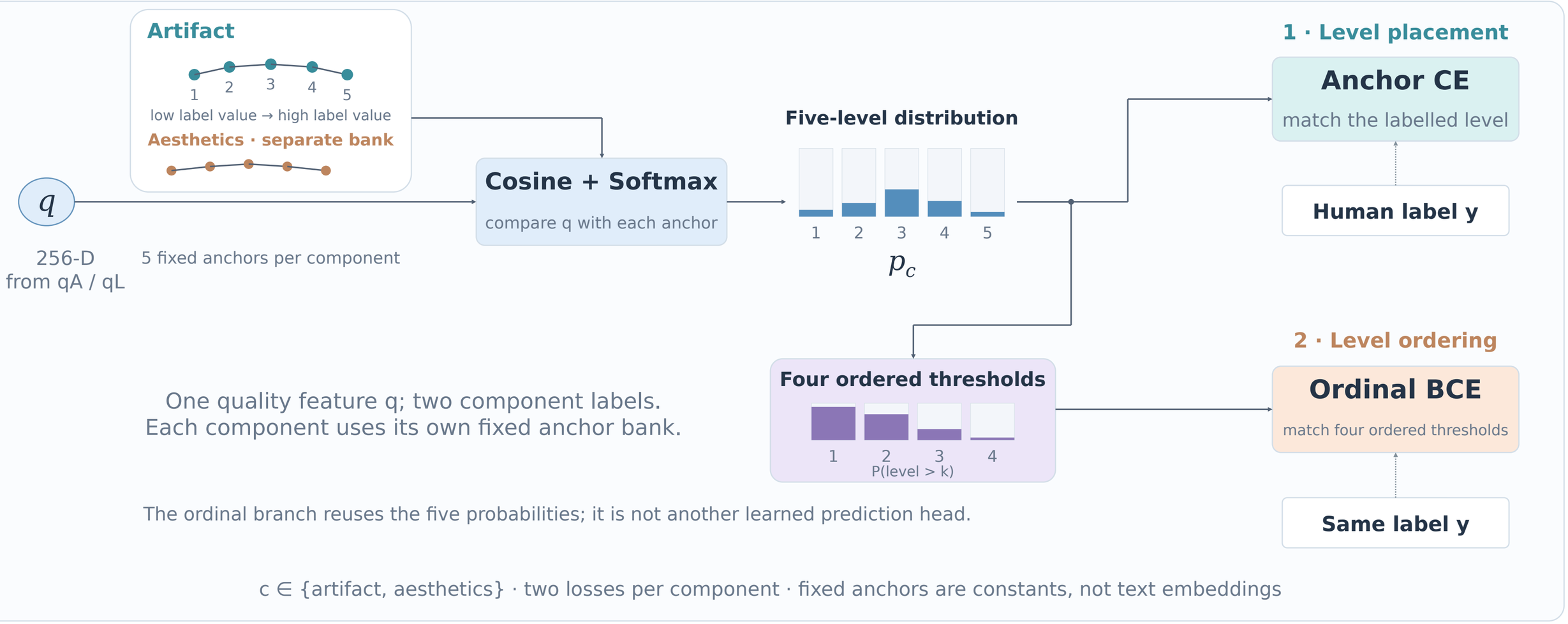
Ordered Component Supervision

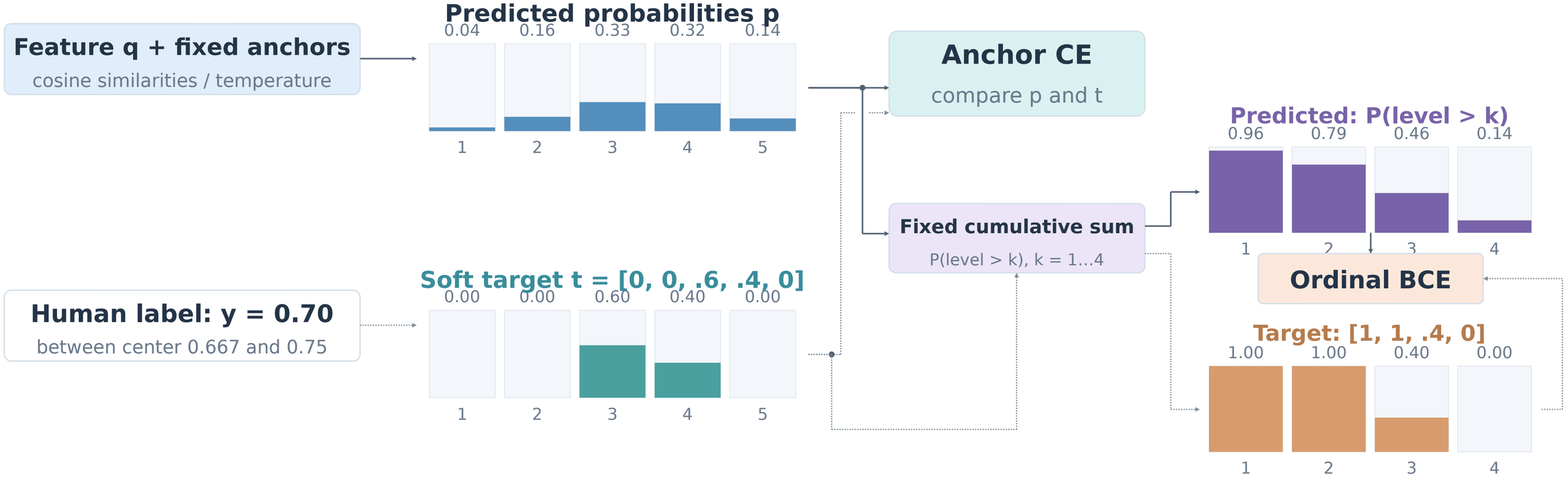
Fixed level anchors shape the representation; cumulative probabilities preserve the level order.



Pretraining only. Both losses send gradients back to q and its global/local representation paths. Bars are schematic; anchors lie in separate planes.

From one human label to two supervision signals

Worked Artifact example: the same five-level distribution supports both anchor and ordinal losses.



Anchor CE asks “Which level?” **Ordinal BCE asks “Which ordered thresholds should be exceeded?”**

The target is generated from the human score. The predictions come from q.
Both losses update the feature extractor; the anchor vectors and cumulative transform remain fixed.

Example values are rounded, follow the ordered-arc / soft-label formulas, and are not experimental results. ε clipping omitted.